

KYLE MASSINGILL, PHD

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National Radio Astronomy Observatory

800 Bradbury SE, Suite 235, Albuquerque, NM 87106

EDUCATION

Ph.D., astrophysics May 2024

Arizona State University

Dissertation: *Water Vapor Millimeter-Wave Instruments and the Regulation of Galaxy Evolution*

(Advisor: Prof. Philip Mauskopf)

B.S., physics May 2018

University of Arizona

EMPLOYMENT HISTORY

National Radio Astronomy Observatory Jun 2024 - Present

Jansky Fellow

Arizona State University School of Earth and Space Exploration Aug 2019 - May 2024

Graduate Researcher

University of Arizona Steward Observatory Jun 2018 - Aug 2019

Research Technician

SKILLS AND EXPERIENCE

Instrumentation

- Led design, simulation, and characterization of a millimeter-wave spectrometer for water vapor radiometry. Created a pipeline for design of resonant cavity waveguide filter-banks, utilizing microwave simulations. Integrated spectrometer on high-altitude balloon platform.
- Development of atmospheric models and radiometric retrieval methods for monitoring phase fluctuations at observatory sites.
- Experienced with CST Studio, *am* atmospheric model, SolidWorks and scikit-rf

Observational Astronomy

- Led continuum stacking analysis of ALMA observed high- z quasars to constrain thermal Sunyaev–Zeldovich halos. Compared stacked observations to feedback and pressure profile models.
- Contributed to a large-scale kinematic study of molecular gas by performing data reduction and imaging of ALMA CO (2-1) observations of $z=1.6$ cluster member galaxies. Utilized baseline phase data to flag and recover ALMA semi-pass data. Performed astrometry correction on HST data.
- Experienced with CASA, SExtractor, SCAMP, and astropy

FELLOWSHIPS

Jansky Fellowship Jun 2024 - Present

National Radio Astronomy Observatory

First Year Fellowship Aug 2019 - Jul 2020

ASU School of Earth and Space Exploration

PUBLICATION LIST

Refereed

- “The SUPERCOLD-CGM Survey. II. [C I](1–0) Emission and the Physical Conditions of Cold Gas in Enormous Ly α Nebulae at $z\sim 2$ ”, J. Li, B. HC. Emonts, Z. Cai, J. Li, R. Wang, M. Villar-Martin, F. A. Battaia, M. Li, Y. Wu, I. Yoon, M. D. Lehnert, **K. Massingill**, C. Sarazin, J. X. Prochaska, M. Lacy, B. Mason, , 2024, The Astrophysical Journal, 977:198 (19pp). (<https://dx.doi.org/10.3847/1538-4357/ad8637>)
- “Observational Constraints on Sunyaev-Zeldovich Effect Halos Around High- z Quasars”, **K. Massingill**, B. Mason, M. Lacy, J. Li, I. Yoon, C. Sarazin, 2024, The Astrophysical Journal, 969, 56. (<https://dx.doi.org/10.3847/1538-4357/ad3a67>)
- “The SUPERCOLD-CGM Survey. I. Probing the Extended CO (4–3) Emission of the Circumgalactic Medium in a Sample of 10 Enormous Ly α Nebulae at $z\sim 2$ ”, J. Li, B. HC. Emonts, Z. Cai, J. Li, F. A. Battaia, J. X. Prochaska, I. Yoon, M. D. Lehnert, C. Sarazin, Y. Wu, M. Lacy, B. Mason, **K. Massingill**, 2023, The Astrophysical Journal, 950, 2. (<https://dx.doi.org/10.3847/1538-4357/acae96>)
- “A large-scale kinematic study of molecular gas in high- z cluster galaxies: Evidence for high levels of kinematic asymmetry”, W. J. Cramer, A. G. Noble, **K. Massingill**, J. Cairns, D. L. Clements, M. C. Cooper, R. Demarco, J. Matharu, M. McDonald, A. Muzzin, J. Nantais, G. Rudnick, H. Übler, E. van Kampen, T. M. A. Webb, G. Wilson, H. K. C. Yee, 2023, The Astrophysical Journal, 944, 2. (<https://dx.doi.org/10.3847/1538-4357/acae96>)
- “Event Horizon Telescope imaging of the archetypal blazar 3C 279 at an extreme 20 microarcsecond resolution”, The Event Horizon Telescope Collaboration et al. 2020, A&A, 640, A69. (<https://doi.org/10.1051/0004-6361/202037493>)
- “First M87 Event Horizon Telescope Results. II. Array and Instrumentation”, The Event Horizon Telescope Collaboration et al. 2019, The Astrophysical Journal Letters, 875, L2, 1-28. (<https://iopscience.iop.org/article/10.3847/2041-8213/ab0c96>)
- “First M87 Event Horizon Telescope Results. I. The Shadow of the Supermassive Black Hole”, The Event Horizon Telescope Collaboration et al. 2019, The Astrophysical Journal Letters, 875, L1, 1-17. (<https://iopscience.iop.org/article/10.3847/2041-8213/ab0ec7>)
- “A VLBI receiving system for the South Pole Telescope”, J. Kim, D. P. Marrone, C. Beaudoin, J. E. Carlstrom, S. S. Doeleman, T. W. Folkers, D. Forbes, C. H. Greer, E. F. Lauria, **K. D. Massingill**, E. Mayer, C. H. Nguyen, G. Reiland, J. SooHoo, A. A. Stark, L. Vertatschitsch, J. Weintraub, A. Young, 2018, Proceedings of SPIE, 10708, 107082S, 1-14. (<http://doi.org/10.1117/12.2301005>)

Selected Conference Proceedings & Abstracts

- “Development of Next Generation Very Large Array Tropospheric Calibration”, **K. Massingill**, T. K. Sridharan, Y. Asaki, 2025, United States National Committee of URSI National Radio Science Meeting.
- “Constraining SZE around high- z Quasars”, **K. Massingill**, B. Mason, M. Lacy, B. Emonts, 2023, American Astronomical Society Meeting #242, 55, 6.
- “CubeSounder: Development of Microwave Radiometer 3D Weather Imaging Sensors”, **K. Massingill**, S. Bryan, P. Mauskopf, C. Groppi, J. Greenfield, M. Baricuatro, D. Carter, R. Suresh, J. Tinlin, D.

Bliss, S. Smas, 2023, Next-generation Suborbital Researchers Conference.

- “Evidence for significant environmental influence on the molecular gas kinematics of cluster galaxies at z 1.6 detected with ALMA”, W. Cramer, A. Noble, **K. Massingill**, J. Cairns, D. Clements, M. Cooper, R. Demarco, J. Matharu, M. McDonald, A. Muzzin, J. Nantais, G. Rudnick, H. Übler, E. van Kampen, T. Webb, G. Wilson, H. Yee, 2023, American Astronomical Society Meeting #241, 55, 2.
- “Millimeter-Wave Filter Bank Spectrometers”, **K. Massingill**, S. Bryan, C. Groppi, P. Mauskopf, B. Pina, P. Rybak, P. Wullen, 2020, Proceedings of the 31st IEEE International Symposium on Space Terahertz Technology.

SEMINARS & PRESENTATIONS

June 2025	Lunch Talk, Green Bank Observatory
May 2025	NRAO PostDoc Symposium, Virtual
April 2025	ISSTT, Berlin Germany
February 2025	Colloquium, NRAO Socorro
January 2025	
	-CART Seminar, University of New Mexico
	-AAS, National Harbor Maryland
	-NRSN, Boulder Colorado
August 2023	Invited Talk, NRAO Socorro
August 2023	Student Talk, NRAO Charlottesville
October 2022	ISSTT (Poster), Baeza Spain
November 2021	Saguaro Astronomy Club, Virtual
March 2021	ASU Engineering Coffee, Virtual

AWARDS

Awarded to the Event Horizon Telescope Collaboration:

Bruno Rossi Prize (High Energy Astrophysics Division of the American Astronomical Society, 2020), **Einstein Medal** (Albert Einstein Society Berne, 2020), **Breakthrough Prize in Fundamental Physics** (Breakthrough Prize Foundation, 2019), **NSF Diamond Achievement Award** (National Science Foundation, 2019)

EDUCATION & OUTREACH

Earth and Space Exploration Open House Committee

Director	2023-2024
Logistics Manager	2022-2023

Substitute Lecturer

SES 126	April 2023
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Teaching Assistant

SES 128: Exploration of the Universe Lab	Spring 2023
SES 123: Earth, Solar System & Universe Lab	Fall 2020

Undergraduate Students Mentored

Ricardo Crespo Vargas (NRAO Summer Student Research Assistantships, 2025)
Michael Baricuatro (ASU Engineering, 2022)
Joe Tinlin (ASU Sundial Project, 2021)
Walter Goodwin (ASU Engineering, 2021)

Jonathan Greenfield (ASU Exploration System Design, 2021)

Jae Joiner (ASU/NASA Space Grant Intern, 2020)

Bianca Pina, Philip Rybak, Peter Wullen (NASA High Altitude Student Platform, 2019)